

P. Shubham Parashar, Ph.D.

Condensed Matter Theory | Quantum Transport | Two-Dimensional Electronic Systems
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Research Profile

Theoretical condensed-matter physicist studying two-dimensional electronic systems. My work develops kinetic, hydrodynamic, electrostatic, and quantum Hall models for disordered and inhomogeneous media, with emphasis on magnetotransport and the theoretical interpretation of nanoscale scanning-probe microscopy experiments in graphene and related heterostructures.

Appointments and Education

2021–2026	Ph.D., Physics , University of California San Diego Advisor: Michael M. Fogler. Dissertation: <i>Electric and Magnetic Response near Perturbations in Two-Dimensional Electron Gases</i> .
2017–2021	Researcher , University of California San Diego Advisors: Benjamin Grinstein and Congjun Wu.
2016–2017	M.S., Physics , Indian Institute of Science, Bangalore Advisor: Rahul Pandit.
2012–2016	B.S., Physics , Indian Institute of Science, Bangalore Advisor: Rahul Pandit.

Publications

Diffusive and hydrodynamic magnetotransport around a density perturbation in a two-dimensional electron gas

P. Shubham Parashar and Michael M. Fogler
Physical Review B **113**, 155443 (2026).

Nanoscale imaging of magnetotransport around a circular p–n junction in graphene

Zachary J. Krebs, Wyatt A. Behn, Keenan J. Smith, Margaret A. Fortman, Kenji Watanabe, Takashi Taniguchi, Pathak S. Parashar, Michael M. Fogler, and Victor W. Brar
Physical Review Letters **136**, 136301 (2026). *Editors' Suggestion; featured in APS Physics Viewpoint.*

Manuscripts and Current Projects

Thermodynamics of the spin-splitting transition in the quantum Hall effect

P. S. Parashar and Michael M. Fogler. Manuscript in preparation.

Tip-induced nonlinear screening and p–n junction formation in graphene

P. S. Parashar and Michael M. Fogler. Manuscript in preparation.

Selected Research Contributions

- Developed analytical and numerical theory of magnetotransport near local density perturbations in two-dimensional electron gases, including diffusive and hydrodynamic regimes.
- Developed and validated semiclassical Boltzmann/Raichev-type transport calculations for scanning tunneling potentiometry near circular graphene p–n junctions.
- Connected analytical asymptotics, numerical parameter scans, and experimentally relevant device parameters in reproducible transport calculations.
- Interpreted electrochemical-potential imaging measurements through local electrostatics, Hall-angle physics, and transport-regime crossovers.

Selected Talks and Presentations

- **Spin-split collapse in higher Landau levels**, UC San Diego, 2023.
- **Vorticity patterns**, Columbia University, 2022.
- **Magnetotransport across local inhomogeneities**, APS March Meeting, 2020.
- **From Berry phases to non-associative operators**, APS March Meeting, 2018.
- **Non-associative operators in SO(5) spin-3/2 electron systems**, UC Irvine, 2018.

Awards and Fellowships

2024	Bhadra Fellowship, UC San Diego
2018–2020	Graduate research support, National Science Foundation
2017	UC San Diego Scholar's Grant
2012–2017	KVPY Fellowship
2015–2016	MEXT-supported research internship, Japan Advanced Institute of Science and Technology

Research Training

2015–2016	Research Intern , Japan Advanced Institute of Science and Technology, Ishikawa Advisor: Ryo Maezono; MEXT-supported.
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Computing

Primary tools	Python; MATLAB; Mathematica/Wolfram Language; LaTeX; Git.
Additional research use	C++; Fortran, used in computational physics projects with Rahul Pandit.
Workflows	Numerical solvers; parameter scans; data analysis; scientific visualization; reproducible computational scripts and notebooks.

References

Michael M. Fogler	Professor of Physics, UC San Diego; mfogler@ucsd.edu.
Rahul Pandit	Honorary Professor; National Science Chair, Indian Institute of Science, Bangalore; rahul@iisc.ac.in.
John McGreevy	Professor of Physics, UC San Diego; mcgreevy@physics.ucsd.edu.